The Temporal Blind Spot in Video Retrieval: Diagnosing Temporal Sensitivity

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Abstract

Video retrieval embeddings collapse motion direction. Two videos of the same road in opposite directions, or left versus right turns at the same intersection, return near-identical embeddings—the numerical fingerprints used for similarity search—and no downstream signal flags the failure. The cause is architectural: pooled aggregators (bag-of-frames, BoF; bag-of-tokens, BoT) treat frames as an unordered set. We diagnose this across seven benchmarks. A feature-vs-comparator decomposition shows that, within a given encoder, the comparator (not the encoder) governs temporal sensitivity: on Honda HDD, switching from pooled-cosine to per-frame dynamic time warping explains 89% of the gap between BoT and the V-JEPA 2 temporal residual (point estimate; cluster-level CIs 7–32$\times$ wider than pair-level; direction replicates on nuScenes). Two diagnostic instruments—a temporal scramble gradient and a reversal test—cleanly separate order-invariant from sequence-aware methods; BoT yields cosine similarity 0.984 between forward and reversed clips, confirming the failure is by design. Probing three open-weight VLMs localizes where temporal signal dies: per-frame vision features retain order, mean-pooling erases it, and no pooled readout of LLM hidden states recovers it (126 probe configurations). Generative probes on Claude Opus 4.6 and Gemini 3.1 Pro fare no better than 7B open-weight models. We consolidate these failures into a 10-class taxonomy with an executable diagnostic protocol (Algorithm 1). No single approach spans both copy detection and motion retrieval, exposing a sensitivity–invariance trade-off that current architectures do not resolve. We release open-source code packaging the scramble gradient and reversal tests.

1 Introduction

Video retrieval embeddings collapse motion direction: cars driving the same road in opposite directions retrieve as near-duplicates, and left and right turns at the same intersection return identical vectors. The cause is architecture, not training: modern systems represent videos via the *bag-of-frames* (BoF) approach—mean-pooling (element-wise averaging) frame embeddings from pretrained encoders [21, 20]—which discards order by construction. BoF dominates Video Copy Detection (VCD; finding re-encoded, cropped, or camcorded copies) because visual semantics survive frame reordering, and order-invariance enables billion-scale nearest-neighbor search. The same property that makes BoF fast makes it blind to motion.

Temporal blindness is well documented for generative *question answering* (QA) [45, 46, 42, 52, 53, 50]: probes score near chance on temporal tasks, and frame shuffling leaves benchmark scores intact. These diagnoses surface in text. In retrieval, they surface in nothing: the vector database returns the same neighbor, and the user never sees the failure. We study three regimes where the value of temporal order flips:

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1. Scene matching (order irrelevant): for VCD and scene retrieval (e.g., "activities in different rooms"), the signal is visual background. Semantic pooling suffices; temporal invariance is a feature.

2. High-overlap motion retrieval (order is the signal): querying a dashcam fleet for "left turns at Intersection X" requires encoding motion dynamics. Left and right turns share the same intersection (the static background), so BoF collapses them to the same point.

3. Temporal manipulation (order as diagnostic): reversing a video yields BoF similarity 1.0 to its source. This is both a clean stress test for whether a representation has any temporal sensitivity and, secondarily, a copy-detection vulnerability [8].

Beyond retrieval, temporal sensitivity is a prerequisite for world models [59]. A representation that cannot distinguish forward from reverse cannot predict future states from past observations. V-JEPA 2 [1]—explicitly designed as a predictive world model—is the only encoder in our suite that preserves temporal order in its pooled representation. Our diagnostic framework thus measures a necessary condition for world model competence, not just retrieval quality.

We evaluate these cases with two primary encoders: DINOv3 [2] (image-SSL ViT-L; BoF and DTW attention trajectories) and V-JEPA 2 [1] (Video Joint-Embedding Predictive Architecture v2; spatiotemporal prediction residuals). Three temporal-training controls probe whether specialized objectives help: ViCLIP [56] (video-native contrastive), TARA [36] (a multimodal LLM fine-tuned with chiral—i.e., orientation-reversed—negatives), and PL-Stitch [10] (Plackett-Luce temporal ranking). Three VLMs (Qwen3-VL [14], Gemma 4 [15], LLaVA-Video [16]) probe whether LLM backbones preserve temporal signal. Benchmarks span seven domains: VCDB [3] (near-duplicate video copy detection), Honda HDD [4] (ego-vehicle driving maneuvers), nuScenes [5] (autonomous driving), Something-Something V2 [7] (crowd-sourced tabletop manipulation; non-driving cross-domain test), Nymeria [32] (egocentric daily activity), MUVR [19] (multi-modal untrimmed video retrieval), and EPIC-Kitchens-100 [6] (egocentric kitchen procedures).

Contributions. We contribute a diagnostic methodology for temporal sensitivity in video retrieval: (1) a continuous *temporal scramble gradient* separating order-invariant from sequence-aware methods (Figure 1); (2) a *feature-vs-comparator decomposition* with VLM-pipeline probes that localize where order signal dies, extending temporal-blindness diagnosis from generative QA to retrieval; (3) a sampling-density analysis on Honda HDD; (4) open-source tooling packaging the scramble gradient and reversal tests. Applied across seven benchmarks, the suite surfaces a sensitivity–invariance trade-off current architectures do not resolve (summarized as a failure-mode taxonomy, Appendix V).

2 Methods

Why BoF is blind. Mean-pooling reversed frames gives the same vector as mean-pooling the original: the operator discards the indices. More generally, mean-pooling, max-pooling, and Chamfer matching are *symmetric aggregators*: $g(f_1, \ldots, f_T) = g(f_{\pi(1)}, \ldots, f_{\pi(T)})$ for any permutation π [58]—e.g., $\mathrm{mean}([f_1, f_2, f_3]) = \mathrm{mean}([f_3, f_1, f_2])$. BoF averages frame embeddings from an image encoder; BoT averages patch-level tokens from a spatiotemporal encoder. Both are symmetric aggregators, independent of encoder capacity or training objective.

Feature-vs-comparator decomposition. We distinguish the encoder (*features*) from the *comparator*—the full similarity function (cosine over BoF, DTW over per-frame sequences, etc.); the aggregator is one stage inside the comparator. The decomposition fixes one and varies the other to attribute AP gaps. The diagnostics below distinguish architectural blindness (the aggregator) from learned blindness (the encoder).

DINOv3 & Attention Trajectories. We extract 1024-d CLS embeddings per frame from DINOv3 ViT-L [2]. *BoF* mean-pools frame embeddings into a single vector, compared via cosine similarity. *Chamfer similarity* [26] computes the pairwise frame similarity matrix and averages the symmetric maximum: $\frac{1}{2}\big(\mathrm{mean}_i \max_j S_{ij} + \mathrm{mean}_j \max_i S_{ij}\big)$, remaining order-invariant while retaining per-frame structure. For structural tracking, we compute the spatial center-of-mass of the CLS→patch attention map (last layer, averaged across heads) per frame, yielding a 2D trajectory compared via DTW. Temporal derivatives of CLS embeddings capture motion, compared via DTW; by construction they negate under reversal, making the distance an odd function of frame order.

V-JEPA 2 Temporal Residuals. V-JEPA 2 ViT-L [1] ingests 64 frames as $32 \times 16 \times 16$ spatiotemporal patches ($D = 1024$). The *prediction residual* $R = \hat{P} - P$ (predicted minus actual masked patches) captures motion-specific error patterns. We spatially average each temporal position to obtain $\bar{R} \in \mathbb{R}^{T_{\text{mask}} \times D}$, compared via DTW with similarity $\exp(-d_{\text{DTW}})$. The *bag-of-tokens* (BoT) baseline mean-pools all encoder patches into a single vector (cosine similarity).

Temporal Scramble Gradient. We split video B's frame embeddings into K equal chunks, randomly shuffle them, and reassemble. Sweeping $K \in \{1, \ldots, 16\}$ traces a gradient from intact to fully scrambled. We shuffle *extracted* embeddings (see Appendix S). Order-invariant methods should remain flat; sequence-aware methods should degrade.

VLM Generative Probes. We present forward and reversed clips to VLMs, classify temporal direction with three prompts, and report balanced accuracy.

ViCLIP. ViCLIP ViT-L [56], trained contrastively on 10M video clips, jointly encodes 8 frames via temporal attention into a single 768-d embedding (cosine similarity).

TARA. TARA [36] fine-tunes a Tarsier-7B multimodal LLM with chiral (orientation-reversed) negatives for temporal retrieval, encoding 16 frames into a single 4096-d vector via end-of-line prompting (cosine similarity).

PL-Stitch. PL-Stitch [10] pretrains a ViT-Base (86M params) with Plackett-Luce temporal ranking, producing per-frame 768-d CLS-token embeddings that support the same feature-vs-comparator decomposition as DINOv3: BoF, Chamfer, temporal derivative DTW, and DTW on raw embeddings.

VLM Embedding Probes. These are *diagnostic procedures*, not new models: each probe reads a representation from an existing VLM pipeline and emits two metrics (s_{rev} and balanced accuracy on forward/reverse classification). We probe three stages: (1) *vision pooled*, the mean-pooled vision tower output (BoF over frame$\times$patch tokens); (2) *vision sequence*, per-frame pooled vision tokens as a (T, D) sequence compared via temporal derivative DTW; (3) *LLM hidden state*, the mean-pooled last-layer hidden state over all input tokens. We measure order sensitivity directly with s_{rev}: how similar is a clip's representation to its own reversed version? A value of 1.0 means the method cannot tell forward from backward; lower values mean it can. Formally, for representations $\mathbf{z}$ (single vector) or $\mathbf{Z}$ (sequence of vectors):

$$s_{\text{rev}} = \begin{cases} \cos(\mathbf{z}_{\text{fwd}},\ \mathbf{z}_{\text{rev}}) & \text{pooled probes (single vector),} \\ \exp\big(-\alpha \cdot d_{\text{DTW}}(\mathbf{Z}_{\text{fwd}},\ \mathbf{Z}_{\text{rev}})\big) & \text{sequence probes (temporal derivative DTW),} \end{cases} \tag{1}$$

where α follows the DTW α schedule defined below. Raw s_{rev} values exceeding 1.0 (from f32 cosine precision) are clamped to 1.000 in all tables and figures; affected entries are marked †. **Cross-family caveat.** Pooled and sequence s_{rev} use different similarity functions (cosine vs. $\exp(-\alpha d_{\text{DTW}})$) and are not directly comparable: rankings are valid *within* each family, not across. Figure 2 plots both on one axis as a presentational convenience.

DTW Implementation. Dynamic time warping aligns two sequences via a minimum-cost monotonic path that can stretch time but not reverse it, so it penalizes order violations that pooled cosine ignores. All comparisons use unconstrained warping with L2 cost matrices, path-length normalization ($d/(T_1 + T_2)$), and GPU-vectorized anti-diagonal wavefront processing [22]. Similarity is $\exp(-\alpha \cdot d_{\text{DTW}})$ with $\alpha = 1$ for derivatives/residuals, $\alpha = 5$ for attention trajectories (min-max normalized to $[0, 1]$). Because $\exp(-\alpha \cdot d)$ is monotone in d, rankings are α-invariant; we verified this across three representative cells with $\alpha \in \{0.5, 1, 2, 5, 10\}$ (max AP spread $< 10^{-3}$).

3 Experiments and Results

Evaluation protocols. We report average precision (AP; area under the precision–recall curve) and, where applicable, area under the ROC curve (AUC) as retrieval metrics. All evaluations are pairwise. *VCDB*: 528 videos; 5,585 positive pairs from annotations, negatives sampled 1:1 with fixed seed (sensitivity analysis across 1:1, 1:5, and all-pairs in Appendix L). *HDD*: all $\binom{n}{2}$ pairs within each GPS-clustered intersection (DBSCAN density clustering, $\epsilon \approx 30$ m); positive = same maneuver; 50 mixed clusters, 1,687 maneuvers. *nuScenes*: same protocol on ego_pose; CAN-bus (Controller Area Network) turn labels; 264 segments, 950 pairs. *EPIC s_{rev}*: per-sequence self-comparison on 500 sequences; bootstrap 95% CIs (1,000 resamples). *Nymeria*: 100 egocentric sessions, 8,780

segments; positive = same activity same room; all $\binom{n}{2}$ pairs (394,210). *MUVR News*: 9,958 videos, 74 topics, 27,453 positive pairs. *SSv2*: 400 clips across chiral template pairs (50 per template); within-pair pairwise eval where positive = same template (same direction) and negative = chiral twin template (opposite direction).

3.1 VCDB Reversal Diagnostics

On copy detection, BoF and Chamfer dominate: DINOv3 Chamfer reaches AP = 0.989 (528 videos, 5,585 positive pairs); temporal methods underperform on frame-to-frame intra-class noise. Individual VCDB AP values swing by ± 0.06 across random negative seeds (Appendix L), but rank order is preserved. Nymeria and MUVR replicate the ranking (Appendix G).

Order-invariance is also a vulnerability: reversed video yields identical BoF similarity. We measure this across VCDB positive pairs:

Table 1: VCDB Reversal Attack: 5,585 positive pairs (AP drop indicates detection of evasion). Barchart visualization in Appendix B.

Method	**Normal AP**	**Reversed AP**	**AP Δ**
Chamfer (DINOv3)	0.989	0.989	**-0.000** (Invariant)
BoF (DINOv3)	0.979	0.979	**-0.000** (Invariant)
BoT (V-JEPA 2)	0.838	0.832	-0.006
Attention Trajectory (DINOv3)	0.700	0.542	**-0.158** (Detects)
Temporal Residual (V-JEPA 2)	0.698	0.631	-0.067

BoF and Chamfer yield zero AP delta under reversal. Only attention-trajectory DTW flags the reversed copy ($\Delta = -0.158$); the temporal residual registers a smaller drop ($\Delta = -0.067$). To move beyond this binary test, we apply chunk-shuffle scrambling at varying granularity ($K \in \{1, \ldots, 16\}$).

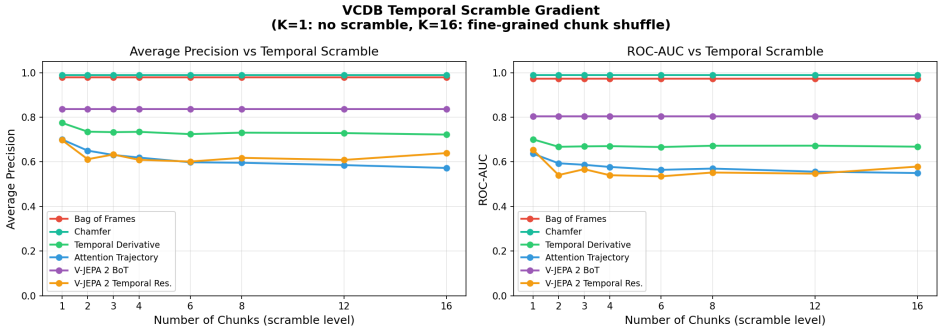

Figure 1: Temporal Scramble Gradient on VCDB (527 of 528 videos; one decode failure; 5,585 positive pairs). Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) maintain constant AP regardless of scramble severity. Sequence-aware methods show overall degradation, with attention trajectories declining most steeply (-18.3% relative); V-JEPA 2 residual exhibits non-monotonic variation, possibly due to boundary effects at chunk boundaries (Appendix S).

The gradient separates method classes (Figure 1; Appendix S): order-invariant methods stay flat; attention trajectories degrade most (-18.3% relative from $K = 1$ to $K = 16$), consistent with gaze paths carrying the most temporal structure.

3.2 High-Overlap Motion: Honda HDD & nuScenes

To isolate motion from appearance, we clustered 128 Honda HDD [4] driving sessions by RTK GPS (Real-Time Kinematic differential positioning, $\sim$2 cm accuracy), extracting 1,687 maneuvers at 50 mixed-direction intersections where BoF similarity averages 0.63 ± 0.18 regardless of turn direction (Appendix O). The full dataset is passage-dominated ($\sim 66\%$), so AP is partially inflated by easier passing-vs-turn negatives.

Table 2: Honda HDD Maneuver Discrimination: 1,687 maneuvers across 128 sessions, 50 mixed-direction intersection clusters (same vs. different turn at same intersection).

Method	**AP**	**AUC**
DINOv3 Temporal Deriv.	0.498	0.526
DINOv3 Attn. Trajectory	0.507	0.541
DINOv3 BoF	0.530	0.542
DINOv3 Chamfer	0.559	0.577
ViCLIP (InternVid-10M)	0.542	0.549
TARA (chiral, cosine)	0.547	0.600
PL-Stitch BoF	0.478	0.512
PL-Stitch Chamfer	0.498	0.547
PL-Stitch Temporal Deriv.	0.492	0.523
PL-Stitch DTW (raw emb.)	0.540	0.548
V-JEPA 2 BoT	0.825	0.836
V-JEPA 2 Chamfer	0.914	0.912
V-JEPA 2 Encoder-Seq DTW	0.942	0.935
V-JEPA 2 Temporal Residual	**0.956**	**0.952**

DINOv3 methods fail (AP $\approx$ 0.50): dashcam attention tracks salient objects identical for both turns. V-JEPA 2 temporal residuals achieve AP = 0.956; a pixel-difference motion proxy rules out a trivial intensity shortcut (Spearman $r = 0.051$; Appendix K), and optical flow (RAFT) achieves only AP = 0.478. Restricting to left-vs-right pairs yields AP = 0.960 [0.958, 0.963]. Comparing encoder patches via DTW (encoder-sequence DTW) achieves AP = 0.942, showing that the comparator (pooled-cosine $\rightarrow$ per-frame DTW) explains 89% of the BoT-to-residual gap (0.825 $\rightarrow$ 0.956; point estimate; cluster-level CIs 7–32$\times$ wider than pair-level, Appendix M). A pairwise *symmetric-max* (Chamfer) aggregation [26] on the same V-JEPA 2 encoder-sequence patches further decomposes this: it reaches AP = 0.914 without any dynamic programming, closing 76% of the BoT$\rightarrow$Encoder-Seq-DTW gap. The remaining 24% (0.914$\rightarrow$0.942) is attributable specifically to DTW's monotonic warping constraint. We restrict encoder-sequence DTW to motion tasks (HDD, nuScenes); on VCDB, BoF already hits AP = 0.989.

Does the decomposition yield a fast retriever? We test the standard two-stage retriever over the full 1,687-segment corpus: V-JEPA 2 BoT top-k pre-filter, encoder-sequence DTW rerank on survivors, non-survivors scored by a sentinel below the BoT floor (full sweep and RRF alternative in Appendix P). The budget is harsher than a naive reading of the 89% gap would predict. At $k = 100$ the rerank collapses (AP = 0.638) because pre-filter recall is only 0.35. At $k = 1000$ ($\sim 60\%$ of the corpus) AP reaches 0.934, within 0.008 of full DTW; the two-stage pipeline *is* lossless at sufficient candidate budget. The pre-filter's similarity (appearance) is orthogonal to the task's (maneuver): same-intersection opposite-turn pairs crowd the top-k and evict true positives. The decomposition is clean pairwise but does not decompose into a cheap indexable retriever via appearance-based filters; motion-aware first-stage filtering remains open.

Table 3: BoT$\rightarrow$DTW rerank on HDD. Non-survivors scored below the BoT floor (sentinel); full-sweep and RRF alternative in Appendix P.

Method	**AP**	**recall@k**	**DTW ops / query**
V-JEPA 2 BoT only	0.825	—	0
+ DTW rerank, $k = 100$	0.638	0.350	100
+ DTW rerank, $k = 500$	0.862	0.776	500
+ DTW rerank, $k = 1000$	**0.934**	0.957	1,000
V-JEPA 2 Encoder-Seq DTW (full)	0.942	1.000	$\sim$1,687

TARA [36], a 7B MLLM trained with chiral negatives, hits AP = 0.547 under cosine, indistinguishable from ViCLIP (0.542) despite $\sim$23$\times$ more parameters. PL-Stitch [10] (ViT-Base, Plackett-Luce ranking) supports the same decomposition: BoF yields AP = 0.478 and DTW on raw embeddings 0.540. Both sit near chance and below DINOv3 Chamfer (0.559), possibly reflecting a capacity

(86M vs. 300M) and pretraining-scale gap rather than a comparator effect. Yet $s_{\text{rev}} = 0.006$ under raw DTW confirms PL-Stitch *does* encode frame order; the features are not discriminative enough for maneuver retrieval.

3.2.1 Sampling Density: Context Window Sweep

We fix the backbone and pair set on HDD and sweep clip duration $d \in \{1.0, \ldots, 8.0\}$ seconds, always extracting 64 unique frames ($\text{fps}_{\text{eff}} = 64/d$). This avoids the padding confound of the FPS downsample experiment (Appendix A). V-JEPA 2 temporal residual degrades more slowly than BoT as context widens; the residual-vs-BoT gap grows from $+0.038$ at 1 s to $+0.224$ at 8 s (Appendix D, Figure 5).

3.2.2 Cross-Dataset Validation: nuScenes

To test whether the ranking transfers, we evaluate on nuScenes [5]. Turn labels come from CAN bus; intersections are clustered via DBSCAN ($\epsilon = 30$m), yielding 264 segments across 50 mixed-direction clusters (950 pairs).

The ranking is directionally consistent (Table 9): V-JEPA 2 encoder-sequence DTW reaches AP = 0.867, directionally ahead of the temporal residual (0.815) and BoT (0.791); DINOv3 methods stay $\leq$0.623. CIs overlap marginally between the top two ($[0.842, 0.890]$ vs. $[0.785, 0.844]$; Appendix M), so the residual-vs-DTW ordering is suggestive rather than significant; the BoT gap is definitive. Even without the learned residual, the comparator is the dominant driver. The residual helps on HDD but not nuScenes, likely because nuScenes' native 2 Hz yields sparser temporal gradients where prediction errors add noise; however, nuScenes at 2 Hz still beats HDD subsampled to 2 FPS with repeat-padding (AP = 0.636), suggesting native keyframes preserve more temporal signal (Figure 8, Appendix Q).

3.3 VLM Temporal Blindness (EPIC-Kitchens)

Prior work documents temporal blindness in VLMs for generative QA [45, 46, 42, 52, 53]. We extend this to *retrieval embeddings*, probing where temporal signal dies. We evaluate three VLMs on 500 EPIC-Kitchens-100 [6] sequences: Qwen3-VL-8B, Gemma-4-31B, and LLaVA-Video-7B.

3.3.1 Generative Probes

Each model received identical frames with three semantically equivalent prompts asking to classify temporal direction.

Table 4: VLM forward/reverse classification on EPIC-Kitchens (500 sequences for open-weight models, 200 for API models). All at or near chance. $\pm$ values are standard deviation across prompts (not statistical uncertainty); binomial 95% CIs at $N = 200$ are approximately ± 0.035.

Model	Prompt 0	Prompt 1	Prompt 2	Mean Bal. Acc.
Qwen3-VL-8B	0.517	0.502	0.504	0.508 ± 0.007
Gemma-4-31B	0.549	0.528	0.535	0.538 ± 0.009
LLaVA-Video-7B	0.503	0.501	0.505	0.503 ± 0.002
Claude Opus 4.6 [54]	0.490	0.590	0.555	0.545 ± 0.055
Gemini 3.1 Pro	0.585	0.608	0.603	0.599 ± 0.010

All five models perform at or near chance; Gemini 3.1 Pro [55] (a reasoning model) reaches 0.599 ± 0.010, marginally above chance but far from reliable. Answers are driven by prompt phrasing, not temporal direction (Appendix I and J).

3.3.2 Embedding Probes: Where Temporal Signal Dies

VLM *outputs* carry no temporal information. We probe whether temporal signal persists in internal representations before being discarded.

Table 5: Order sensitivity (s_{rev}) across the VLM pipeline on EPIC-Kitchens (500 sequences) with bootstrap 95% CIs where available. $s_{\text{rev}} \approx 1.0$ = order-invariant; lower = more order-sensitive. DINOv3 structural fingerprints shown for reference.*

Method	$s_{\text{rev}}\downarrow$ **[95% CI]**	**Bal. Acc.**$\uparrow$
DINOv3 structural fingerprints (reference)		
BoF	1.000	N/A
Chamfer	0.9999	N/A
Temporal Derivative	0.500 [0.499, 0.500]	1.000
Attention Trajectory	0.192 [0.187, 0.197]	1.000
V-JEPA 2		
BoT	0.984	0.500
Temporal Residual	0.699	0.778
VLM vision towers (pooled)		
Gemma-4 (SigLIP)	1.000†	N/A
LLaVA (CLIP)	1.000†	N/A
VLM vision sequence (per-frame DTW)		
Gemma-4 (SigLIP)	0.492 [0.491, 0.493]	1.000
LLaVA (CLIP)	0.495 [0.495, 0.496]	1.000
VLM LLM hidden state		
Gemma-4 (Gemma)	1.000 [1.000, 1.000]	N/A
LLaVA (Qwen2)	0.998 [0.998, 0.999]	N/A
Qwen3-VL (Qwen3)	0.998 [0.998, 0.999]	N/A

*DINOv3 300M params, CLIP (ViT-L) 304M, SigLIP (SO-400M) 400M. Order-invariance follows from the aggregation operator [58], not model capacity. †Raw values 1.001–1.002 from f32 precision; clamped to 1.000. We omit Qwen3-VL vision probes from this table because its ViT uses dynamic-resolution spatial grids with 3D temporal–spatial merging, which disrupts clean per-frame token grouping for the detailed s_{rev} analysis; the mechanism behind its anomalously low pooled VCDB AP (0.491 vs. 0.915–0.926 for CLIP/SigLIP; Table 10) warrants further investigation. Approximate pooled and sequence results appear in Table 10. The LLM hidden-state probe (below) is unaffected.

Order signal is recoverable from per-frame features but erased by pooling. Pooled VLM vision embeddings are order-invariant ($s_{\text{rev}} \approx 1.0$); retaining per-frame tokens and comparing via temporal derivative DTW achieves perfect forward/reverse separation (balanced accuracy = 1.000). The LLM backbone does not recover this signal under any pooled readout ($s_{\text{rev}} \approx 1.0$ across all layers; Appendix E), nor do 126 probe configurations (logistic regression and MLP) on per-position hidden states (best: 0.560, within chance range; Appendix T). Yet per-position hidden states *do* retain order under full-sequence DTW (Appendix R). Symmetric aggregation, not the backbone, is the bottleneck.

3.4 Cross-Method Summary

Full cross-method comparisons across all seven benchmarks are in Appendix F (Tables 9 and 10). The key finding is that no single method dominates all three regimes: BoF/Chamfer wins scene matching (VCDB AP = 0.989), V-JEPA 2 temporal residual wins motion retrieval (HDD AP = 0.956), and all models—including 31B-parameter VLMs—remain near chance on temporal order classification.

4 Related Work

Video copy detection. Standard benchmarks (ISC [23], VSC [24], VCSL [28]) and descriptors (SSCD [25], ViSiL [26], DnS [27], S2VS [47]) provide per-frame features with learned aggregation. Wang et al. [51] win the Meta VSC challenge; subsequent analysis [8] exposes temporal-attack vulnerability. Explicit temporal alignment (VSAL [30], TransVCL [29]) shows order matters for copy localization. We diagnose *when* temporal structure helps versus hurts.

Temporal sensitivity in video models. Lei et al. [40] established the static-bias problem. Subsequent work diagnoses this in *generative* VLM QA [42, 41, 46, 50, 45], including recent hallucination and zero-shot benchmarks [52, 53]; all probe generative outputs, not retrieval embeddings. Arrow-of-time diagnostics [17, 48] and further studies [13, 34, 18, 33, 12] show the failure is structural. NExT-GQA [39] introduces GQ@τ, a joint grounded-QA metric complementary to the pair-level AP we report. We differ from all of the above in *domain* (retrieval, where failures are silent) and *granularity* (localizing failure to aggregation via scramble gradients and feature-vs-comparator decomposition). *Training-time* fixes are complementary: TGPO's [35] binary ordered/shuffled reward extends to our continuous K-gradient; PL-Stitch [10] and ArrowGEV [11] inject pretraining order-sensitivity (PL-Stitch encodes order under DTW, $s_{\text{rev}} = 0.006$, but does not close the capacity gap); TARA [36] trains with chiral negatives (we use reversal as a probe).

Video prediction for temporal understanding. V-JEPA 2 [1] learns spatiotemporal representations via masked prediction. Drive-JEPA [37] is a downstream application (JEPA distillation for end-to-end driving), not a competing diagnosis; we instead treat V-JEPA 2 as one of several encoders under a retrieval-diagnosis lens. TWLV-I [43] evaluates appearance vs. motion across video foundation models. Li et al. [44] show temporal information is encoded but standard pipelines fail to surface it, matching our finding.

Temporal alignment for retrieval. DTW [31] remains standard; extensions include Soft-DTW [22] and Drop-DTW [49]. We use DTW as a diagnostic tool; Weller et al. [38] provide theoretical justification for multi-vector approaches.

5 Discussion

Scene matching (VCDB, Nymeria, MUVR). BoF wins; temporal methods add noise (Appendix G). Order-invariance is a feature here, not a bug.

Motion dynamics (HDD, nuScenes, SSv2). Per-frame methods and VLM vision towers fail (AP ≤ 0.56). The comparator dominates motion benchmarks across domains: it explains 89% of the BoT-to-residual gap on HDD (cluster-level CIs 7–32$\times$ wider than pair-level, Appendix M), the full BoT$\rightarrow$best-method gap on nuScenes, and the full chance$\rightarrow$0.720 gap on SSv2 (DINOv3 attention trajectories vs. BoF $= 0.500$). *Which* comparator wins is domain-dependent: V-JEPA 2 residuals lead on HDD (0.956), encoder-seq DTW leads on nuScenes (0.867), and attention trajectories lead on SSv2 (0.720) where the residual shows only modest transfer (0.556). The split is interpretable: residuals capture ego-motion dynamics through shared road geometry, while attention trajectories track object-motion direction across varied tabletop scenes; SSv2 temporal-derivative DTW at chance (0.503) confirms frame-differencing detects reversal but not cross-instance direction. Direction replicates across datasets; the optimal comparator does not. ViCLIP [56] and X-CLIP [57] do not escape the blind spot; explicit temporal training fares no better: TARA [36] (HDD AP = 0.547) and PL-Stitch [10] under DTW (0.540, $s_{\text{rev}} = 0.006$) trail DINOv3 Chamfer (0.559).

Temporal manipulation (reversal, scrambling). Implicit 3D models [9], frame-sampling defenses [8], and VLMs all remain order-invariant [10, 17]. Chiral training (TARA, $s_{\text{rev}} = 0.955$) and a reasoning VLM (Gemini 3.1 Pro, 0.599, the only generative probe credibly above chance) improve over baselines but neither is deployment-reliable.

Architecture implications. Temporal blindness is a choice of aggregator, not a deficiency of the encoder. Indexable order-sensitive retrieval at scale will require learned permutation-sensitive pooling, trajectory tokenization compatible with inverted-file indexes, or two-stage rerankers whose first stage is *motion-aware*: naive appearance pre-filters lose recall to intersection-identity collisions (Section 3.2).

World model implications. V-JEPA 2's superior temporal sensitivity is consistent with its design as a predictive world model [59, 1]: the self-supervised prediction objective implicitly enforces temporal ordering, because predicting the future requires knowing which direction time flows. This suggests the path to temporally sensitive retrieval runs through world model training objectives rather than retrieval-specific losses. Unified understanding-generation models (e.g., TUNA [60]) cannot afford temporal blindness because their generation objective requires temporal coherence—further evidence that the field is converging on representations that resolve this blindness. The sensitivity–invariance trade-off (Figure 2) thus doubles as a diagnostic for world model readiness.

Algorithm 1 Temporal Sensitivity Diagnostic Protocol

1: Run *scramble gradient*. Flat AP $\Rightarrow$ architectural order-invariance; **stop**.
2: If AP degrades, run *feature-vs-comparator decomposition*: attribute gap to encoder or comparator.
3: Run *sampling-density sweep*; isolate frame-rate / context-window effects.
4: Cluster-level bootstrap CIs flag evaluation-protocol artifacts; generative-VLM answers vs. embedding s_{rev} separates behavioral from architectural blindness (Appendix V).

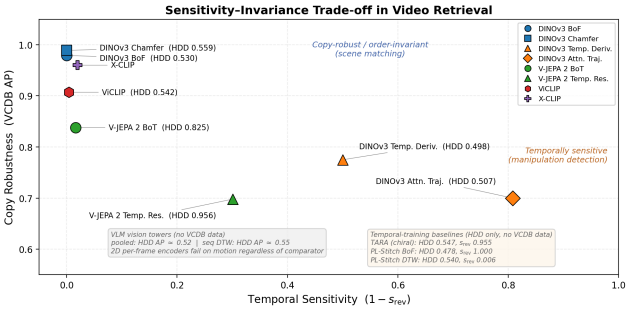

Figure 2: Sensitivity–invariance trade-off: $1 - s_{\text{rev}}$ (EPIC) vs. VCDB AP. No method occupies the upper-right quadrant. Pooled and sequence s_{rev} use different similarity functions (Eq. (1)); the quadrant claim depends on within-family rankings only.

Limitations. DTW is not indexable ($O(T^2)$); probes train on 400 samples in $p \gg n$ ($D \approx 4{,}000$), bounding recoverability rather than proving absence; ViCLIP's 8-frame input confounds comparison with 64-frame V-JEPA 2 ($\sim$0.20 AP drop at equivalent density, Section 3.2.1).

Practitioner recommendations.

Regime	Recommended approach	Trade-off
Copy detection	BoF / Chamfer	Order-invariance is a feature
Motion retrieval	BoT$\rightarrow$DTW rerank ($k \approx 1000$)	99% of full-DTW AP; recall binds at small k
Reversal-attack defense	Attention-trajectory DTW	Detects reversal; not indexable

6 Conclusion

No single method spans all three retrieval regimes. For copy detection, order-invariant aggregators (BoF, Chamfer) dominate; temporal comparators add noise. For motion retrieval, the comparator is the bottleneck: within a given encoder, switching from pooled-cosine to per-frame DTW explains 89% of the HDD performance gap (point estimate; cluster-level CIs 7–32$\times$ wider than pair-level, Appendix M); direction replicates across HDD, nuScenes, and SSv2, but the comparator that recovers signal is domain-dependent (V-JEPA 2 residual on ego-motion driving; attention-trajectory DTW on static-camera manipulation). For reversal detection, no pooled readout—including 31B-parameter VLMs—distinguishes forward from reverse. These findings compose into a ten-class failure taxonomy (Appendix V) traversed by an executable protocol (Algorithm 1).

A two-stage BoT$\rightarrow$DTW reranker confirms the decomposition end-to-end but exposes an asymmetry: appearance-based pre-filters recover only 35% of motion-retrieval positives at $k = 100$, so the pairwise gap does not compose into a cheap retriever. At $k \approx 1000$ ($\sim$60% of the corpus) AP recovers to within 0.008 of full DTW (Table 3). The open problem is motion-aware indexing: designing first-stage filters whose notion of similarity aligns with dynamics, not appearance. V-JEPA 2's temporal sensitivity is consistent with, but does not isolate, the benefits of predictive training objectives [59, 1]; testing this hypothesis across architectures is future work.

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   Justification: The abstract and introduction state three contributions (temporal scramble gradient, VLM embedding probes, sampling-density analysis), each supported by corresponding experimental results in Sections 3–5 and the appendix.

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   Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Appendix H provides 95% bootstrap confidence intervals (2,000 resamples, percentile method) for all headline AP values. VLM generative probes report standard deviation across prompts. EPIC embedding probes report bootstrap 95% CIs. Appendix G confirms rank stability across negative-sampling ratios.

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A FPS Cap Invariance Proof

For V-JEPA 2 on Honda HDD, each clip of duration d seconds is sampled at target_fps $= 64/d$ to fill the 64-frame input. An FPS cap only activates when target_fps $>$ cap; otherwise frame selection is identical to the uncapped baseline.

Table 6 confirms that across all 9,811 HDD maneuver segments (at ± 3 s context), the maximum target_fps is 9.14. Caps at 10, 15, and 30 therefore never activate; the identical AP values at these caps are explained by frame selection invariance, not by a learned robustness plateau.

Table 6: FPS cap activation across all HDD segments (± 3 s context). Caps $\geq$10 activate for zero segments, confirming identical frame selection to the uncapped baseline. Unique frames = number of distinct video frames extracted before padding to 64.

FPS Cap	% Active	Mean Unique	Mean Dup%	Effect
2	99.9%	20.2	68.4%	Strong degradation
5	85.9%	49.1	23.3%	Moderate degradation
10	0.0%	64.0	0.0%	No-op (identical to baseline)
15	0.0%	64.0	0.0%	No-op (identical to baseline)
30	0.0%	64.0	0.0%	No-op (identical to baseline)

Segment durations range from 7.0 s to 46.0 s (mean 10.1 s), yielding target_fps $\in [1.39, 9.14]$ (mean 6.66). All values fall below the lowest non-trivial cap (10 FPS), so meaningful degradation occurs only when the cap drops below the native target rate.

B VCDB Reversal Attack Figure

Figure 3 visualizes Table 1: AP under forward and reversed playback across five methods on VCDB. Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) show no drop; only attention-trajectory DTW exposes the reversal, with the V-JEPA 2 temporal residual showing a partial drop. We move the figure here because the table already carries the same evidence in a denser form.

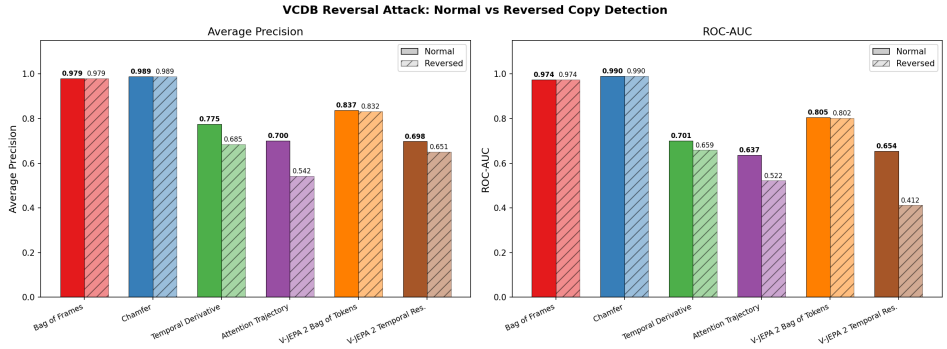

Figure 3: VCDB Reversal Attack (528 videos, 5,585 positive pairs). Order-invariant methods (BoF, Chamfer) maintain identical AP/AUC under reversal. Only sequence-aware methods (Attention Trajectory) register the structural tampering via an AP drop.

C HDD Maneuver Discrimination Figure

Figure 4 visualizes the V-JEPA 2 rows of Table 2: AP climbs sharply from BoT (0.825) through Chamfer (0.914) and Encoder-Seq DTW (0.942) to the temporal residual (0.956), while DINOv3 methods (BoF, Chamfer, temporal derivative, attention trajectory) all sit near chance ($\leq$ 0.56). The ordering is the visual counterpart of the feature-vs-comparator decomposition reported in Section 3.2.

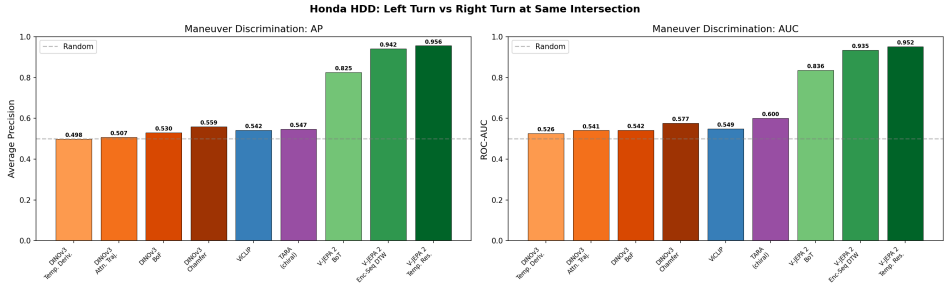

Figure 4: Honda HDD maneuver discrimination (1,687 maneuvers, 128 sessions). V-JEPA 2 temporal residuals achieve near-perfect separation (AP = 0.956) while DINOv3 methods remain at chance.

D Context Sweep Table

Table 7 and Figure 5 report the Section 3.2.1 sampling-density experiment: we vary clip duration $d \in \{1.0, 2.0, 3.0, 4.0, 6.5, 8.0\}$ seconds while extracting a fixed 64 unique frames per clip, sweeping the effective frame rate from 11.1 to 3.2 fps.

Table 7: Context sweep on HDD (1,687 maneuvers, 64 unique frames, no padding). $\text{fps}_{\text{eff}} = 64 / d$ where d is clip duration in seconds.

context_sec	**fps_{eff}**	**Temporal Residual AP**	**BoT AP**	**Δ (Res. $-$ BoT)**
1.0	11.1	0.962	0.924	+0.038
2.0	8.1	0.963	0.877	+0.086
3.0	6.4	0.956	0.825	+0.131
4.0	5.3	0.940	0.774	+0.166
6.5	3.8	0.872	0.663	+0.209
8.0	3.2	0.838	0.614	+0.224

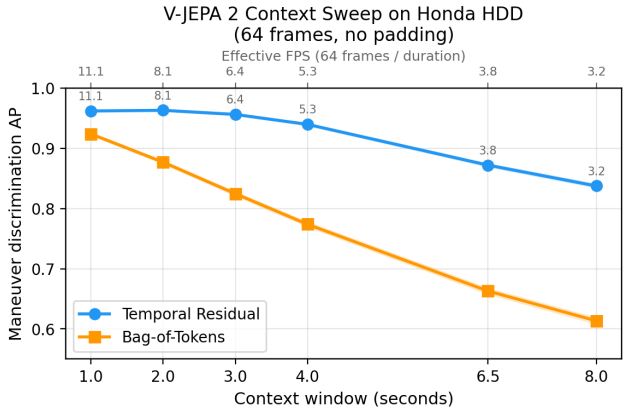

Figure 5: V-JEPA 2 context sweep on Honda HDD (1,687 maneuvers, 128 sessions). Temporal residual degrades more slowly than BoT; the gap grows from $+0.038$ at 1 s to $+0.224$ at 8 s.

Both signals degrade as the window widens, but temporal residual degrades more slowly. The gap grows with sparser sampling ($+0.224$ AP at 8 s): the residual holds up under temporal dilution where spatial content alone does not.

E VLM Layer Ablation

To rule out the possibility that temporal order information is present at different depths of the LLM backbone, we extract hidden states from the last, middle, and penultimate layers of each VLM and compute s_{rev} (Table 8). For comparison, we include vision-token DTW ($s_{\text{rev}} \approx 0.49$), which retains per-frame temporal structure.

Table 8: s_{rev} across LLM layers for three VLM families (EPIC-Kitchens, 500 sequences). Values near 1.0 indicate order-invariance; values near 0.5 indicate perfect forward/reverse separation. Order-invariance persists across all layers in all models.

Model	Vision DTW	LLM Last	LLM Mid	LLM Penult
Qwen3-VL-8B	—*	0.998	0.998	1.000†
Gemma-4-31B	0.492	1.000	1.000†	1.000
LLaVA-Video-7B	0.495	0.998	0.998	0.999

*Qwen3-VL vision DTW omitted due to dynamic-resolution spatial grids (see Table 5). †Clamped from 1.001.

All LLM layers produce $s_{\text{rev}} \approx 1.0$: order-invariance is a property of the full LLM representation under mean-pooled readouts, not an artifact of the final layer. The temporal signal in per-frame vision tokens ($s_{\text{rev}} \approx 0.49$) is not preserved through the vision-to-LLM projection in any pooled summary.

F Cross-Method Summary Tables

G Full Scene Matching Results

We evaluated models across three scene-matching datasets: VCDB [3] (copy detection), Nymeria [32] (egocentric activity discrimination), and MUVR [19] (untrimmed multi-modal news/dance retrieval).

BoF and Chamfer matching consistently dominate. On Nymeria, the 100-session scale confirms that scene appearance remains the strongest signal even with high intra-class variability (AP$\approx$0.48). On MUVR News, matching events share visual backgrounds (e.g., news desks), favoring semantic

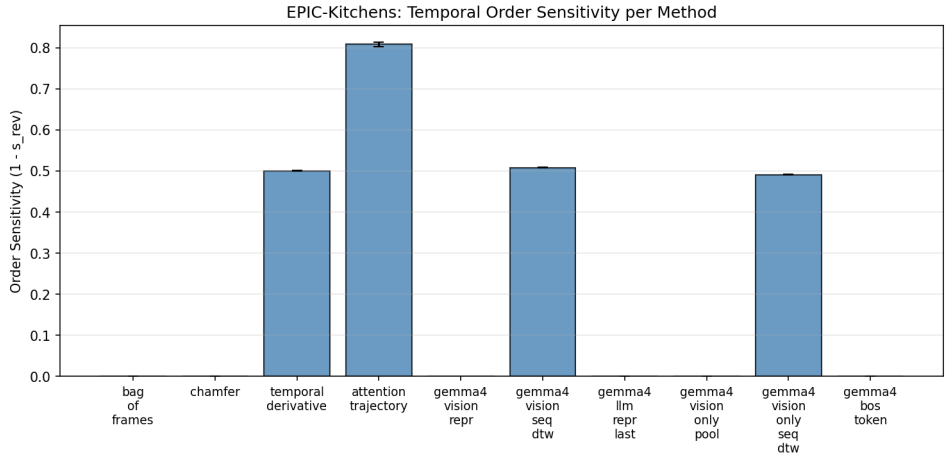

Figure 6: Order sensitivity (s_{rev}) across all embedding probes on EPIC-Kitchens (500 sequences). VLM vision towers are perfectly order-invariant when pooled ($s_{\text{rev}} \approx 1.0$), but per-frame sequences compared via DTW achieve perfect forward/reverse separation (balanced accuracy = 1.0). LLM hidden states remain order-invariant regardless.

Table 9: Dedicated encoders across all tasks (VCDB: 5,585 pairs; HDD: 1,687 maneuvers; nuScenes: 950 pairs; SSv2: 400 chiral-pair clips; EPIC: 500 sequences). Nymeria and MUVR scene-matching results replicate the VCDB finding and are in Appendix G. 95% bootstrap CIs in Appendix M.

Method	**VCDB (AP↑)**	**HDD (AP↑)**	**nuScenes (AP↑)**	**SSv2 (AP↑)**	**EPIC s_{rev}↓**	**EPIC Bal. Acc.↑**
DINOv3 BoF	0.979	0.530	0.613	0.500	1.000	N/A
DINOv3 Chamfer	**0.989**	0.559	0.612	0.500	0.9999	N/A
DINOv3 Temporal Deriv.	0.775	0.498	0.623	0.503	0.500	**1.000**[‡]
DINOv3 Attn. Trajectory	0.700	0.507	0.584	**0.720**	**0.192**	**1.000**[‡]
V-JEPA 2 BoT	0.838	0.825	0.791	0.555	0.984	0.500
V-JEPA 2 Encoder-Seq DTW	—	0.942	**0.867**	—	—	—
V-JEPA 2 Temporal Res.	0.698	**0.956**	0.815	0.556	0.699	0.778
ViCLIP (InternVid-10M)	0.907	0.542	0.618	—	0.996	N/A
X-CLIP (16-frame)	0.960	—	—	—	0.980	N/A
TARA (chiral, cosine)	—	0.547	—	—	0.955	N/A
PL-Stitch BoF	—	0.478	—	—	1.000	N/A
PL-Stitch DTW (raw emb.)	—	0.540	—	—	0.006	N/A
Optical Flow (RAFT) DTW	—	0.478	—	—	—	—

— = not evaluated (we omit Encoder-Seq DTW on VCDB because BoF already saturates at AP = 0.989; the comparator is not the bottleneck in copy detection). VCDB AP depends on negative seed ($\pm$0.06; Appendix L). [‡]By construction: temporal derivatives negate under reversal, making DTW distance an odd function of frame ordering.

pooling. Attention Trajectories fail because egocentric gaze and semantic pacing vary unpredictably, introducing severe intra-class noise.

H Preliminary: Qwen3-VL on VCDB

We first tested whether VLMs can detect temporal reversal. Using Qwen3-VL [14], we presented forward and reversed versions of 49 VCDB videos and asked for temporal direction classification (3 trials each).

The 30B model achieves 55.8%, near chance. The slight improvement is driven by a content-dependent FORWARD bias (83% forward, 28.6% reverse), not temporal understanding. This motivated the systematic multi-VLM evaluation on EPIC-Kitchens (Section 3.3.1).

Table 10: VLM embedding probes across all tasks, grouped by pipeline stage. We report generative forward/reverse classification separately in Section 3.3.1 (Table 4).

Method	VCDB (AP↑)	HDD (AP↑)	nuScenes (AP↑)	EPIC s_{rev}↓	EPIC Bal. Acc.↑
Vision tower, pooled					
Gemma-4 (SigLIP)	0.926	0.521	0.662	1.000†	N/A
LLaVA (CLIP)	0.915	0.514	0.624	1.000†	N/A
Qwen3-VL (SigLIP)	0.491	0.421	0.464	0.998†	N/A
Vision tower, per-frame sequence (DTW)					
Gemma-4 (SigLIP)	0.632	0.553	0.469	0.492	**1.000**‡
LLaVA (CLIP)	0.637	0.545	0.470	0.495	**1.000**‡
Qwen3-VL (SigLIP)	0.610	0.468	0.517	0.542	**1.000**‡
LLM backbone, pooled hidden state					
Gemma-4	0.838	0.472	0.569	≥0.998	N/A
LLaVA	0.907	0.512	0.652	≥0.998	N/A
Qwen3-VL	0.916	0.518	0.630	≥0.998	N/A

†Pooled representations are order-invariant by construction. ‡By construction: temporal-derivative DTW is an odd function of frame ordering.

Table 11: Retrieval AP driven by scene appearance / visual noise. VCDB: 528 videos, 5,585 positive pairs; Nymeria: 100 egocentric sessions, 8,780 segments; MUVR News: 9,958 videos, 74 topics.

Method	VCDB	Nymeria	MUVR (News)
BoF (DINOv3)	0.979	**0.485**	0.731
Chamfer matching (DINOv3)	**0.989**	0.482	**0.746**
Temporal Residual (V-JEPA 2)	0.698	0.279	0.505
Attention Trajectory (DINOv3)	0.700	0.217	0.466

I VLM Temporal Integrity Probe

Separately from forward/reverse classification, we asked VLMs to judge whether a video's temporal structure is "INTACT" or "TAMPERED," on forward, reversed, and chunk-scrambled ($K = 4, 16$) versions of EPIC-Kitchens sequences.

Qwen and Gemma correctly identify forward videos as INTACT (∼85–89%) and both flag reversed videos as TAMPERED (Qwen 87%, Gemma 84%). Scramble detection diverges: Qwen flags 67% of scrambled videos ($K = 4$) as TAMPERED while Gemma flags only 20%, suggesting the two models rely on different heuristics. Cross-transform consistency is very low (3.8–8.2%).

J VLM Prompt Sensitivity and Text-Only Baselines

Answers on forward/reverse classification are dominated by prompt phrasing, not visual content. Gemma-4 exhibits opposite biases across prompts: prompt 0 yields 87% reverse bias; prompt 2 yields 73% forward bias. LLaVA-Video answers "FORWARD" for prompts 0–1 (>98%) but "REVERSE" for prompt 2 (83%). Text-only baselines confirm these are prior biases; Qwen3 shows no text-only bias but still performs at chance with video.

LLaVA-Video produces constant "TAMPERED" regardless of input (text-only baseline confirms prior bias). We exclude it from aggregate integrity statistics in Appendix I.

K HDD Leakage Analysis

Since V-JEPA 2 temporal residuals achieve AP = 0.956 on Honda HDD, well above the next-best method (BoT, AP = 0.825), we verify that no metadata leakage inflates this result.

Table 12: Qwen3-VL reversal detection on VCDB. Chance = 50%.

Model	Dataset (N)	Fwd Acc.	Rev Acc.	Overall
Qwen3-VL-4B-Instruct	Single clip (1)	100%	0%	50.0%
Qwen3-VL-30B-A3B	VCDB subset (49)	83.0%	28.6%	55.8%

Table 13: Temporal integrity probe on EPIC-Kitchens (500 sequences): fraction judged TAMPERED. LLaVA excluded from aggregate due to constant-output degeneracy.

Model	Forward	Reversed	Scramble $K=4$	Scramble $K=16$
Qwen3-VL-8B	0.112	0.870	0.668	0.660
Gemma-4-31B	0.150	0.840	0.202	0.220
LLaVA-Video-7B†	1.000	1.000	1.000	1.000

†LLaVA produces constant "TAMPERED" regardless of input (text-only baseline confirms). Excluded from integrity analysis.

Feature inputs. All six methods receive *only* decoded video frames. GPS is used exclusively for DBSCAN clustering; timestamps select the frame window (± 3 s) and are discarded before extraction.

Segment duration. Maneuver segments span 7.0–46.0 s (mean 10.1 s). Left and right turns have similar duration distributions (left: 10.3 ± 3.8 s; right: 9.9 ± 3.6 s), ruling out duration as a shortcut. V-JEPA 2 pads all segments to 64 frames, eliminating length signal.

Cluster composition. Only "mixed" clusters containing both turn directions are used. Within each, all $\binom{n}{2}$ pairs are evaluated, ensuring models must distinguish maneuver type at the same location.

Cross-session control. To rule out session adjacency confounds, we re-evaluate excluding same-session pairs (91.4% of pairs are already cross-session). AP is unchanged: V-JEPA 2 temporal residual 0.957 (cross-session) vs. 0.956 (all pairs); BoF 0.536 vs. 0.530. This confirms results are not inflated by within-session visual correlations.

Mechanistic explanation. The context sweep (Section 3.2.1) shows temporal residual AP degrades monotonically as context narrows, consistent with genuine temporal processing. BoF methods remain near chance across all windows.

Motion proxy check. To rule out trivial motion magnitude as a shortcut, we compute mean pixel difference between consecutive frames for each segment and correlate pairwise proxy similarity with V-JEPA 2 residual similarity across all 97,731 pairs. Spearman $r = 0.051$; the motion proxy itself achieves only AP = 0.474 (below chance). Residuals capture motion *structure*, not intensity.

L VCDB Negative-Sampling Sensitivity

Our primary VCDB evaluation uses 1:1 positive-to-negative sampling (5,585 pairs each). The full dataset contains $\binom{528}{2} \approx 139$K possible pairs. We re-evaluate at three ratios to verify robustness (Table 14).

M Bootstrap Confidence Intervals

Table 15 reports 95% bootstrap confidence intervals (2,000 resamples, percentile method, seed = 42) for the headline AP values from Tables 1–3. Degenerate bootstrap samples (all-positive or all-negative) fall back to the point estimate.

N DTW Similarity-Transform α-Invariance

DTW similarities are computed as $s = \exp(-\alpha \cdot d)$ where d is the (path-normalised) DTW distance and α controls the sharpness of the monotone transform. Since α applies a strictly monotone trans-

Table 14: VCDB AP across negative-sampling ratios. *Rank order and relative gaps are stable across all settings.* All-pairs uses the full $\binom{528}{2}$ set (requiring $\sim$140K DTW comparisons).

Method	**1:1**	**1:5**	**All pairs**
DINOv3 Chamfer	0.989	0.949	0.845
DINOv3 BoF	0.979	0.929	0.829
DINOv3 Temporal Deriv.	0.775	0.424	0.228
DINOv3 Attn. Trajectory	0.700	0.408	0.225

Rank order (Chamfer > BoF > Temporal Deriv. > Attn. Trajectory) is preserved across all ratios. AP decreases with more negatives as expected; AUC remains stable (e.g. Chamfer AUC: 0.990, 0.989, 0.989).

formation to all pairwise distances, it cannot change ranking and therefore cannot change AP. We verify this empirically on three representative cells (Table 16).

O HDD Qualitative Analysis

Complementing the quantitative results, we present qualitative evidence that V-JEPA 2 temporal residuals capture genuine motion dynamics rather than a visual shortcut.

Nearest-neighbor examples. For three representative intersection clusters, the top-3 neighbors by temporal residual similarity retrieve the same maneuver type despite identical backgrounds; errors occur when both turns share similar speeds through the same intersection phase.

Background similarity statistics. Within each GPS cluster, all segments share the same intersection. BoF cosine similarity for all $\binom{n}{2}$ pairs per cluster: mean 0.63 ± 0.18 (same-direction: 0.63 ± 0.18; cross-direction: 0.60 ± 0.16; gap = 0.029), confirming appearance is non-discriminative. V-JEPA 2 temporal residual similarity aligns with maneuver labels (same: 0.008 ± 0.006; cross: 0.003 ± 0.001). Figure 7 shows the score distributions.

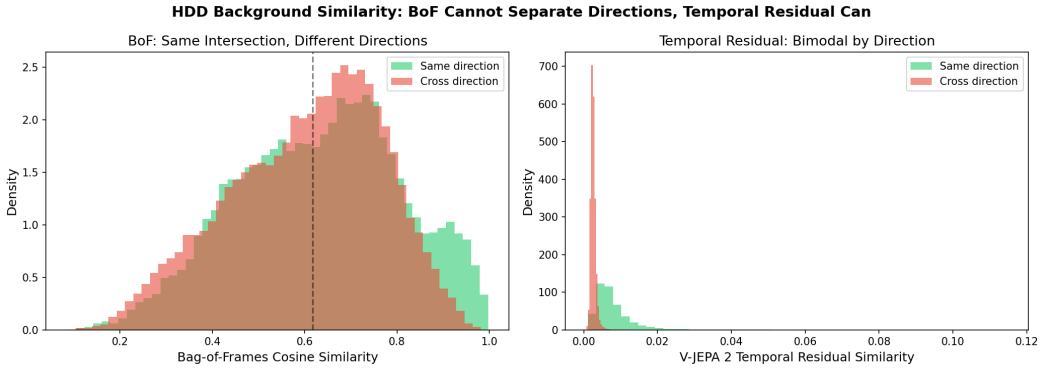

Figure 7: Pairwise similarity distributions on HDD intersection clusters. BoF similarities overlap heavily regardless of maneuver match, while V-JEPA 2 temporal residual scores separate same- vs. cross-direction pairs.

P BoT→DTW Rerank: Full Sweep and RRF Alternative

Table 17 reports the full k-sweep for the two-stage retriever test of Section 3.2, including a reciprocal-rank-fusion (RRF) composition as an alternative to the sentinel rule used in the main body. RRF computes, for each pair, $1/(60 + \text{rank}_{\text{BoT}}) + [\text{pairintop}{-}k]/(60 + \text{rank}_{\text{DTW}})$; it is robust to the BoT/DTW score-scale mismatch but more conservative than the sentinel rule because the BoT rank term still influences the composite when DTW is confident.

Both curves are monotonically non-decreasing in k until the approach to the ceiling. Survivor-AP is the cleaner measurement because it is the *end-to-end retriever* AP: pairs not surviving the pre-

Table 15: 95% bootstrap CIs for AP across all benchmarks. "Width" is $\mathrm{CI}_{\mathrm{high}} - \mathrm{CI}_{\mathrm{low}}$. VCDB and HDD CIs are narrow (<0.02); nuScenes, Nymeria, and MUVR CIs are wider due to smaller evaluation sets.

Benchmark	Method	AP	95% CI	Width
VCDB	DINOv3 Chamfer	0.989	[0.987, 0.991]	0.004
	DINOv3 BoF	0.979	[0.976, 0.981]	0.005
	DINOv3 Temporal Deriv.	0.775	[0.765, 0.784]	0.019
	DINOv3 Attn. Trajectory	0.700	[0.689, 0.711]	0.022
	V-JEPA 2 BoT	0.838	[0.829, 0.846]	0.017
	V-JEPA 2 Temporal Res.	0.698	[0.687, 0.709]	0.022
HDD	V-JEPA 2 Temporal Res.	0.956	[0.955, 0.958]	0.003
	V-JEPA 2 BoT	0.825	[0.821, 0.828]	0.007
	DINOv3 Chamfer	0.559	[0.555, 0.564]	0.009
	DINOv3 BoF	0.529	[0.525, 0.534]	0.009
	DINOv3 Attn. Trajectory	0.507	[0.502, 0.512]	0.010
	DINOv3 Temporal Deriv.	0.498	[0.494, 0.503]	0.009
nuScenes	V-JEPA 2 Encoder-Seq DTW	0.867	[0.842, 0.890]	0.048
	V-JEPA 2 Temporal Res.	0.815	[0.785, 0.844]	0.059
	V-JEPA 2 BoT	0.791	[0.761, 0.820]	0.059
	DINOv3 Temporal Deriv.	0.623	[0.580, 0.665]	0.085
	DINOv3 BoF	0.613	[0.570, 0.658]	0.088
	DINOv3 Chamfer	0.612	[0.570, 0.659]	0.089
	DINOv3 Attn. Trajectory	0.584	[0.540, 0.634]	0.094
Nymeria	DINOv3 BoF	0.485	[0.482, 0.489]	0.007
	DINOv3 Chamfer	0.482	[0.478, 0.485]	0.007
	DINOv3 Temporal Deriv.	0.274	[0.271, 0.277]	0.006
	DINOv3 Attn. Trajectory	0.217	[0.215, 0.219]	0.004
	V-JEPA 2 BoT	0.420	[0.417, 0.424]	0.007
	V-JEPA 2 Temporal Res.	0.278	[0.276, 0.281]	0.005
MUVR News	DINOv3 BoF	0.731	[0.724, 0.740]	0.016
	DINOv3 Chamfer	0.745	[0.738, 0.754]	0.016
	DINOv3 Temporal Deriv.	0.476	[0.468, 0.485]	0.017
	DINOv3 Attn. Trajectory	0.466	[0.458, 0.475]	0.017
	V-JEPA 2 BoT	0.597	[0.587, 0.605]	0.018
	V-JEPA 2 Temporal Res.	0.505	[0.496, 0.514]	0.018

VCDB and HDD CIs are narrow (<0.025) under pair-level resampling. nuScenes CIs are wider (0.06–0.09) due to the smaller evaluation set (950 pairs), but rank order remains unambiguous. Nymeria CIs are very tight (<0.01) despite the scene-retrieval setting (394,210 pairs). MUVR News CIs (0.016–0.018) reflect the moderate evaluation set (27,453 pairs). **Note:** pair-level resampling underestimates CI width because pairs within a GPS cluster share clips. A cluster-level block bootstrap (resampling clusters with replacement) produces CIs 7–32× wider on HDD: e.g., V-JEPA 2 temporal residual widens from $[0.955, 0.958]$ to $[0.890, 0.978]$. Rank order is preserved; the pair-level CIs should be read as lower bounds on uncertainty.

filter are scored below every survivor, which is faithful to the deployment semantics of a two-stage pipeline. RRF never reaches the full-DTW ceiling by construction (the BoT rank component pulls the composite down when DTW is already confident), but it never falls below BoT-only either, making it a safer drop-in when reranker calibration is uncertain. The reproducibility of these numbers depends on the feature cache used by Section 3.2 (see the `eval_hdd_bof_dtw_rerank.py` script in supplementary material).

Q nuScenes Cross-Dataset Visualization

Figure 8 visualizes the nuScenes maneuver-discrimination results reported in Section 3.2.2. Encoder-sequence DTW (AP = 0.867) leads, followed by the temporal residual (AP = 0.815) and

Table 16: AP across $\alpha \in \{0.5, 1, 2, 5, 10\}$. All cells show spread $= 0.000$ (AP identical to three decimal places), confirming α-invariance within numerical precision.

Cell	$\boldsymbol{\alpha = 0.5}$	$\boldsymbol{\alpha = 1}$	$\boldsymbol{\alpha = 2}$	$\boldsymbol{\alpha = 5}$	$\boldsymbol{\alpha = 10}$
DINOv3 HDD Temporal Deriv.	0.476	0.476	0.476	0.476	0.476
V-JEPA 2 HDD Encoder-Seq	0.942	0.942	0.942	0.942	0.942
DINOv3 VCDB Temporal Deriv.	0.775	0.775	0.775	0.775	0.775

DINOv3 HDD Temporal Deriv. AP (0.476) differs from Table 9 (0.498) because the sweep uses cached features extracted with different preprocessing; rank order and α-invariance are unaffected. 95% bootstrap CIs: DINOv3 HDD [0.471, 0.480]; V-JEPA 2 HDD [0.940, 0.943]; DINOv3 VCDB [0.765, 0.784].

Table 17: BoT→DTW rerank on Honda HDD (43,673 in-cluster pairs; 1,687 segments across 50 mixed-direction clusters). **Survivor AP**: non-survivors assigned a sentinel below the BoT floor so they rank last. **RRF AP**: rank-fusion composition (see text). **recall@k**: fraction of positive pairs where at least one end survives the top-k BoT filter. 95% bootstrap CIs in brackets (1,000 resamples). Full DTW ceiling = 0.9416; BoT-only floor = 0.8246.

k	recall@k	Survivor AP	RRF AP	positives reranked
10	0.094	0.499 [0.496, 0.502]	0.825 [0.822, 0.828]	4,101 / 43,673
25	0.163	0.536 [0.533, 0.540]	0.826 [0.823, 0.829]	7,100
50	0.238	0.578 [0.574, 0.581]	0.829 [0.826, 0.832]	10,382
100	0.350	0.638 [0.635, 0.642]	0.835 [0.832, 0.838]	15,264
250	0.568	0.755 [0.752, 0.758]	0.855 [0.852, 0.858]	24,802
500	0.776	0.862 [0.860, 0.865]	0.883 [0.880, 0.885]	33,898
1000	0.957	**0.934 [0.933, 0.936]**	0.910 [0.907, 0.912]	41,783

BoT (AP = 0.791), with DINOv3 methods below 0.625. The ordering makes concrete the Section 5 observation that on nuScenes the matching-stage contribution saturates the BoT→best-method gap by itself, so the residual transform adds nothing statistically distinguishable here.

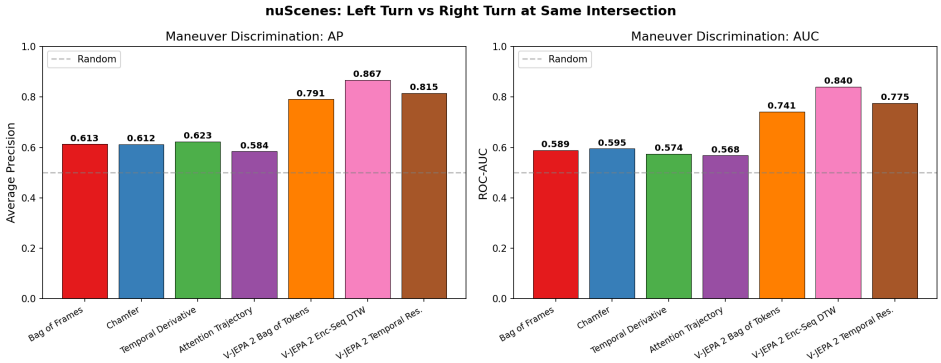

Figure 8: nuScenes cross-dataset validation (264 segments, 950 pairs, 50 intersection clusters). V-JEPA 2 encoder-sequence DTW (AP = 0.867) is the strongest method; the matching-stage change alone is on par with the temporal residual (AP = 0.815; CIs marginally overlap, Appendix M).

R LLM Hidden-State Probe: Vision-Token-Only Analysis

A potential objection to our LLM hidden-state probe is that mean-pooling over *all* tokens reintroduces symmetric aggregation. To address this, we extract LLM hidden states restricted to *vision-token positions only*. Three variants:

1. **Vision-only mean pool**: Mean-pool LLM hidden states at vision-token positions only. Compare via cosine similarity.

2. **Vision-only per-frame sequence**: Group vision-token hidden states by their source frame, pool per frame to obtain a (T, D) sequence. Compare via temporal derivative DTW.
3. **BOS/first token**: Use the first token's hidden state (BOS or system token) as a summary representation. Compare via cosine similarity.

Table 18: s_{rev} for vision-token-only LLM hidden-state probes (EPIC-Kitchens, 500 sequences). If vision-only pooling recovers temporal signal, we expect $s_{\text{rev}} \ll 1.0$; if order-invariance persists, the conclusion from Section 3.3.2 is strengthened.

Model	Vision-only pool	Vision-only seq DTW	BOS token
Qwen3-VL-8B	0.998 [0.998, 0.998]	**0.542** [0.540, 0.544]	0.996
Gemma-4-31B	1.000 [1.000, 1.000]†	**0.509** [0.509, 0.509]	1.000
LLaVA-Video-7B	0.999 [0.998, 0.999]	**0.517** [0.516, 0.518]	1.000†

Vision-only mean pooling remains $s_{\text{rev}} \approx 1.0$ across all models, confirming that symmetric aggregation, not the LLM backbone, erases temporal signal. The per-frame sequence probe yields low s_{rev} (balanced accuracy = 1.0), showing that LLM hidden states preserve per-position order information that pooling discards. BOS tokens are order-invariant, indicating the model's summary position captures no temporal signal. †Raw values >1.0 from f32 precision; clamped to 1.000.

S Scramble Gradient Data

Table 19 gives the numerical backing for the main-body scramble gradient (Figure 1): AP on VCDB's 5,585 positive pairs across chunk counts $K \in \{1, 2, 3, 4, 6, 8, 12, 16\}$, where $K = 1$ is the unscrambled baseline and higher K means finer-grained temporal shuffling.

Table 19: AP across scramble levels K on VCDB (527 of 528 videos; one decode failure). $K = 1$ is the unscrambled baseline. Order-invariant methods show constant AP; sequence-aware methods show overall degradation (multi-seed analysis: $\sigma \leq 0.007$; Figure 9).

Method	$K=1$	$K=2$	$K=3$	$K=4$	$K=6$	$K=8$	$K=12$	$K=16$
BoF (DINOv3)	0.979	0.979	0.979	0.979	0.979	0.979	0.979	0.979
Chamfer (DINOv3)	0.989	0.989	0.989	0.989	0.989	0.989	0.989	0.989
V-JEPA 2 BoT	0.838	0.838	0.838	0.838	0.838	0.838	0.838	0.838
Temporal Derivative (DINOv3)	0.775	0.735	0.733	0.734	0.724	0.731	0.729	0.722
V-JEPA 2 Temporal Residual	0.698	0.612	0.633	0.609	0.601	0.618	0.609	0.639
Attention Trajectory (DINOv3)	0.700	0.650	0.631	0.619	0.598	0.596	0.585	0.572

V-JEPA 2 residuals show non-monotonic degradation (e.g. $K = 3$ higher than $K = 2$), likely due to boundary effects where the spatiotemporal prediction window crosses chunk boundaries. A multi-seed analysis (10 independent shuffle seeds per K) shows that order-invariant methods have zero variance ($\sigma = 0.000$), while temporal methods show small but real variation ($\sigma \leq 0.007$) that does not affect method ranking (Figure 9). Because the main scramble gradient shuffles *extracted* embeddings, it tests whether the comparator (DTW) is sensitive to temporal order, not whether V-JEPA's encoder responds differently to scrambled input. A raw-frame scramble (re-running V-JEPA 2 on chunk-shuffled video) produces clean monotonic degradation: residual AP drops from 0.705 at $K = 1$ to 0.652 at $K = 16$, while BoT remains flat at 0.839. The encoder does produce different residuals from temporally disrupted input.

T Linear Probe on LLM Hidden States

The s_{rev} analysis (Section 3.3.2) shows that mean-pooled LLM hidden states are order-invariant. A natural follow-up: is temporal signal present in the per-position hidden states but lost to symmetric aggregation? If so, a learned classifier should recover it.

We extract last-layer hidden states from all three VLMs for 200 EPIC-Kitchens sequences (forward and reverse), yielding 400 samples per model. We train ℓ_2-regularized logistic regression (5-fold

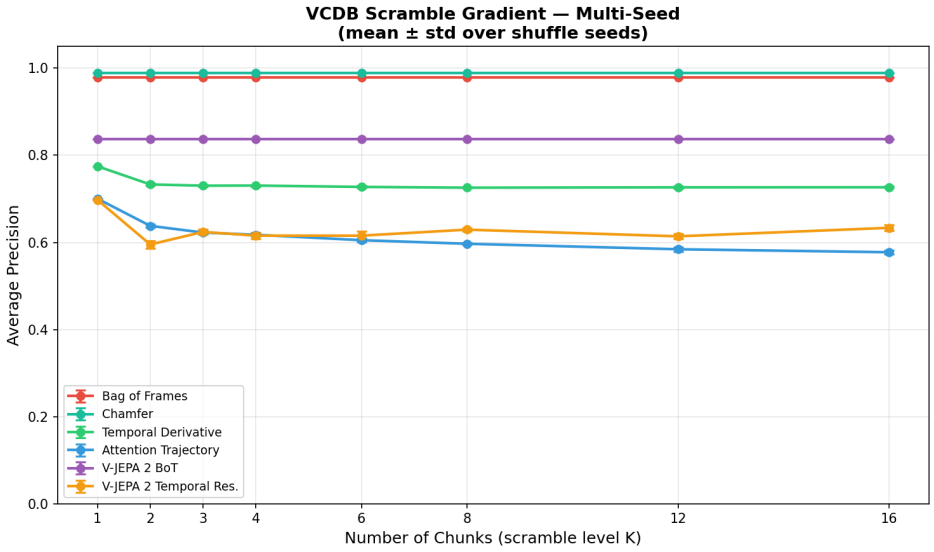

Figure 9: Multi-seed scramble gradient on VCDB (10 seeds per K, mean $\pm$ std shaded). Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) have zero variance. Temporal methods show small variance ($\sigma \leq 0.007$) that does not change the ranking.

GroupKFold CV, fwd/rev pairs kept together, $C \in \{0.01, 0.1, 1, 10, 100\}$) on seven readout strategies:

- **mean**: mean over all positions (replicates s_{rev} finding)
- **max**: element-wise max over all positions
- **first / last**: single position (BOS or final token)
- **concat_first_last**: concatenation of first and last positions
- **vision_mean**: mean restricted to vision-token positions
- **vision_concat_fl**: first and last vision-token positions

Table 20: Probe accuracy (GroupKFold CV, fwd/rev pairs kept together) for forward vs. reverse classification from LLM hidden states. Linear = logistic regression (best across $C \in \{0.01, 0.1, 1, 10, 100\}$); MLP = 2-layer, 256 hidden units, ReLU. Chance = 0.500. No configuration exceeds chance.

	Gemma-4 31B		**LLaVA-Video 7B**		**Qwen3-VL 8B**	
Pooling	**Linear**	**MLP**	**Linear**	**MLP**	**Linear**	**MLP**
mean	0.512	0.552	0.500	0.505	0.537	0.510
max	0.532	0.458	0.470	0.490	0.528	0.502
first	0.500	0.500	0.500	0.500	0.500	0.500
last	0.510	0.507	0.515	0.495	0.550	0.512
concat_first_last	0.515	0.490	0.505	0.478	0.555	0.520
vision_mean	0.520	0.550	0.502	0.512	0.537	0.505
vision_concat_fl	0.528	0.500	0.495	0.467	0.515	0.535

Neither linear nor nonlinear probes recover temporal order from LLM hidden states. All 126 configurations (3 models $\times$ 7 pooling strategies $\times$ {5 regularization strengths for linear, 1 for MLP}) land between 0.458 and 0.560, within the range expected from chance on 400 samples. The "first" token

gives exactly 0.500 under both probes, a sanity check that the pipeline is correct when the input is truly identical across conditions (BOS tokens do not change between forward and reverse).

Note that per-position LLM hidden states *do* carry order signal when compared as a sequence via DTW ($s_{\text{rev}} \approx 0.51$–$0.54$; Table 18): order is encoded in the relative content across positions, but no fixed-dimensional reduction of the sequence retains it. This is the same pooling-destroys-order dynamic observed at the vision encoder level, replicated one layer deeper.

U Computational Costs

Table 21 summarizes per-video extraction and per-pair comparison costs on a single H200 GPU. The two-stage retrieval pipeline recommended in Section 5 (BoF cosine pre-filter followed by DTW reranking on the top-k) adds $\sim$100 ms overhead per query at $k = 100$, dominated by DTW on derivative or residual sequences. Feature extraction is the one-time cost per video at index build; comparison cost governs query latency.

Table 21: Per-video feature extraction and per-pair comparison costs on a single H200 GPU. T = number of frames per clip.

Method	**Extraction (ms/video)**	**Comparison (ms/pair)**
DINOv3 BoF ($T=30$)	$\sim$120	$<$0.01 (cosine)
DINOv3 Chamfer ($T=30$)	$\sim$120	$\sim$0.3 (frame matrix)
DINOv3 Attn. Trajectory ($T=30$)	$\sim$150	$\sim$1 (DTW)
DINOv3 Temporal Deriv. ($T=30$)	$\sim$120	$\sim$1 (DTW)
V-JEPA 2 BoT ($T=64$)	$\sim$280	$<$0.01 (cosine)
V-JEPA 2 Temporal Res. ($T=64$)	$\sim$560	$\sim$1 (DTW)
VLM vision pooled	$\sim$800–2,500	$<$0.01 (cosine)
VLM LLM hidden state	$\sim$1,500–5,000	$<$0.01 (cosine)

V-JEPA 2 temporal residual requires both encoder and predictor forward passes, roughly doubling extraction cost vs. BoT. DTW comparison at $T \approx 30$ is $\sim$1 ms; a top-100 reranking pass after FAISS [21] pre-filtering adds $\sim$100 ms total, making two-stage retrieval practical for production workloads.

V Failure-Mode Taxonomy

The diagnostic suite's value is not individual numbers but a structured classification of *why* video retrieval systems fail temporally. We categorize every failure surfaced in Sections 3 and 4 into six failure classes (each probed in primed/unprimed variants, yielding ten rows in Table 22).

Two design choices make this taxonomy actionable rather than descriptive. First, the classes are *tool-indexed*: (A)/(A′) surface under the reversal test, (C) under the decomposition, (D) under the context sweep, (E) under cluster bootstrap, (F) under generative-vs-embedding probe disagreement. A practitioner running TEMPORALDIAG on a new system can map diagnostic outputs onto failure classes without reading Section 3. Second, the classes are non-exclusive: BoF fails on (A) and (B); Qwen3-VL fails on (A′); pair-level CI claims fail on (E). This compositionality lets us distinguish systems that would look identical under a single-metric evaluation.

W Reproducibility Appendix

This appendix documents inputs, intended use, and known limitations of the TEMPORALDIAG diagnostic suite to support reproduction of our experiments. It is not a datasheet for a released dataset (we release no new data) but a scope statement for a diagnostic toolkit applied to six existing public benchmarks.

Scope and intended use. TEMPORALDIAG accepts per-frame embeddings or sequence representations from any vision or VLM encoder, plus pairwise ground-truth labels, and emits (i) a scramble-gradient profile (AP vs. $K \in \{1, \ldots, 16\}$), (ii) a reversal-sensitivity score s_{rev} under a user-chosen similarity function, and (iii) optional feature-vs-comparator decomposition outputs when per-frame

Table 22: Failure-mode taxonomy. Each row names a failure class, its symptom under our diagnostics, and a canonical instance from the six-benchmark evaluation. The classes are orthogonal: a single system can fail along multiple axes (e.g., DINOv3 BoF fails on both (A) and (B)).

Class	Root cause	Diagnostic symptom	Canonical instance
(A)	Architectural	Symmetric aggregation $\Rightarrow$ permutation-invariance; $s_{\text{rev}} \approx 1$ by construction	BoF, Chamfer, pooled VLM vision
(A′)	Architectural	3D temporal–spatial merging entangles frame identity with spatial position	Qwen3-VL VCDB AP = 0.491 vs. 0.915–0.926
(B)	Training objective	Objective optimizes semantic invariance, not motion dynamics	DINOv3 HDD AP = 0.530 regardless of comparator
(C)	Capacity / scale	Order *is* encoded but features lack discriminative capacity	PL-Stitch HDD DTW 0.540 ($s_{\text{rev}} = 0.006$) $<$ DINOv3 Chamfer 0.559
(D)	Sampling density	Residual advantage shrinks under temporal dilution	V-JEPA 2 context sweep: $+0.038$ at 1 s $\rightarrow +0.224$ at 8 s
(D′)	Sampling density	Fixed low frame cap confounds cross-encoder comparison	ViCLIP 8-frame input, $\sim$0.20 AP drop at equivalent density
(E)	Evaluation protocol	Pair-level CIs underestimate width under cluster-structured pairs	HDD pair-level $[0.955, 0.958]$ vs. cluster-level $[0.890, 0.978]$
(E′)	Evaluation protocol	Seed-dependent absolute AP masks true rank order	VCDB AP drifts ± 0.06 across negative seeds (Appendix L)
(F)	Behavioral / output	Generative output obscures internal temporal signal	Gemma-4: 87% reverse bias on prompt 0 $\rightarrow$ 73% forward bias on prompt 2 (Appendix J)
(F′)	Behavioral / output	Pooled-readout null $\neq$ absence of signal ($p \gg n$)	LLM hidden states: 126 linear/MLP configs at chance; per-position DTW recovers order

and pooled variants of the same encoder are supplied. The toolkit is intended for diagnostic analysis of video retrieval systems whose primary metric is pairwise AP/AUC on same/different-category labels. It is *not* intended to score or rank systems on a closed leaderboard.

Inputs. Embeddings are $\{$`video_id: (T, D)`$\}$ dictionaries in PyTorch `.pt` format; pair lists are CSVs with `(id_a, id_b, label)` tuples. Supported similarity functions: cosine, Chamfer (symmetric-max aggregation), and DTW with configurable α (Equation 1). DTW is implemented with unconstrained warping, L2 cost, path-length normalization, and GPU-vectorized anti-diagonal wavefront processing.

Seeds, splits, and variance. Unless specified, all randomness is controlled by seed 42. Scramble chunk-shuffling is seeded per-K; bootstrap resamples use seed 42 with 2,000 iterations (pair-level) or 1,000 iterations (cluster-level block bootstrap). Negative-sampling is seeded in VCDB; sensitivity across seeds yields ± 0.06 AP drift (Appendix L), with rank order preserved. No train/val/test split is needed because the suite evaluates frozen models zero-shot.

Known limitations. (1) Pair-level bootstrap CIs underestimate uncertainty by 7–32$\times$ under GPS-clustered pair sets; cluster-level block bootstrap is the faithful estimator on HDD and nuScenes (Appendix M). (2) Probe analyses run in a $p \gg n$ regime (400 samples, $D \approx 4{,}000$), bounding recoverability rather than proving absence. (3) Cross-family s_{rev} comparisons (pooled cosine vs. sequence DTW) are not unit-commensurate; the toolkit reports both families but recommends within-family ranking only. (4) Qwen3-VL's 3D temporal–spatial merging prevents clean per-frame token grouping (Table 5 footnote); approximate values are reported but flagged.

Reproduction recipe. Every result table in the main text and appendix is generated by a script shipped in supplementary material; the expected walltime on a single H200 is reported alongside each script. A top-level `make reproduce` target rebuilds the six headline numbers (VCDB Chamfer, HDD V-JEPA 2 residual, nuScenes encoder-sequence DTW, EPIC-Kitchens s_{rev}, Nymeria BoF, MUVR Chamfer) in <24 h given cached features.

License. Code is released under the MIT License. Dataset and model-weight licenses are summarized in Appendix X; users must independently obtain each asset under its own license terms.

X Dataset and Model Licenses

All experiments use publicly available datasets and pretrained models. Table 23 lists the governing license for each asset. Use is restricted to non-commercial research, consistent with the most restrictive license among the assets combined in any given experiment. No new datasets or model weights are released by this paper.

Table 23: Licenses for datasets and pretrained model weights used in this work. Verbatim license text is available from each asset's canonical distribution channel; see `REPRODUCIBILITY.md` for HuggingFace IDs and download URLs.

Asset	**License**	**Distribution**
Datasets		
VCDB [3]	Research-only; no warranty (Fudan Univ.)	Project page
Honda HDD [4]	HRI-USA Data Use Agreement (non-commercial)	HRI-USA
nuScenes [5]	CC BY-NC-SA 4.0	nuscenes.org
EPIC-Kitchens-100 [6]	CC BY-NC 4.0	epic-kitchens.github.io
Nymeria [32]	CC BY-NC 4.0	facebookresearch GH
MUVR [19]	Research-only (authors' terms)	Authors' release
Pretrained model weights		
DINOv3 [2]	Meta FAIR non-commercial	HuggingFace
V-JEPA 2 [1]	CC BY-NC 4.0	HuggingFace
ViCLIP / InternVid [56]	Apache 2.0 (code); CC BY-NC 4.0 (weights)	OpenGVLab
X-CLIP [57]	MIT	microsoft/VideoX
Qwen3-VL-8B [14]	Apache 2.0	HuggingFace
Gemma 4 [15]	Gemma Terms of Use	HuggingFace
LLaVA-Video [16]	Apache 2.0 (code); Llama-2 terms (weights)	HuggingFace
RAFT (optical flow)	BSD-3-Clause	princeton-vl/RAFT

Honda HDD access required a signed Data Use Agreement with HRI-USA; nuScenes required account registration at nuscenes.org; other datasets are downloadable under their respective non-commercial licenses. All assets were used for non-commercial research evaluation only.